Derivation of Intertemporal Targets for Large Pumped Hydro Energy Storage With Stochastic Optimization

J. P. Deane, E. J. McKeogh, and B. P. Ó. Gallachóir

***Abstract*—This paper models large pumped hydro energy storage in a future power system where variable generation, primarily in the form of wind generation, is the dominant source of power generation. The research question posed is how to formulate day-ahead and week-ahead reservoir targets for pumped hydro energy storage in the context of wind forecast uncertainty. The innovation in the work is the use of historical wind data series and wind forecasts to derive a management strategy for the operation of large PHES using stochastic optimization that outperforms current methods in power systems with significant wind generation. This approach derives intertemporal targets for large pumped hydro energy storage that reduce overall system costs when compared to targets derived using a conventional method.**

***Index Terms*—Pumped hydro energy storage, stochastic optimization.**

I. Introduction

THE extent to which any power system can accommodate variations in electrical supply is governed to a large degree by its flexibility. Pumped hydro energy storage (PHES) is often cited as a potential technical facilitator for large scale wind energy integration due to its inherent flexibility, quick start characteristics and its ability to store excess energy [1]. During the next 8 years, over 7 GW of PHES may be added to the European network [2] with a predicted doubling of PHES capacity within Europe by 2050 [3] indicating a renewed interest in this technology. As wind penetration levels increase, curtailment, in addition to other power system issues, is likely to become more prominent [4]. By storing electrical energy when renewable output is high and the demand low, and generating when renewable output is low and the demand high, the curtailment of RE will be reduced, and the baseload units on the system will operate more efficiently [5], [6]

In this paper, the modeling of large PHES in a power system with significant wind penetration is addressed and in particular the challenges associated with the determination of end of day and end of week targets for large energy storages in an environment with wind forecast uncertainty. Two important background assumptions for this work are 1): The problem is addressed from a transmission system operator (TSO) point of view and not from a market perspective. The TSO is mandated with the safe, secure, reliable, economical and efficient operation of the transmission and power system. 2) Large closed-loop PHES is considered, this means that the energy capacity of the storage is large. A number of these large systems are proposed worldwide [7], [8].

Section II provides a review of current modeling techniques while Section III presents the modeling methodology used in this analysis and the test system. Section IV presents the results and Section V draws conclusions.

II. Pumped Hydro Energy Storage Modeling

In contrast with thermal generation, whose operation is decoupled in time (except large baseload plant with high startup/shutdown cost), PHES systems are coupled in time in that a decision today affects operating costs in the future. For example the decision to release or hold water today may inhibit or enhance the ability to capture and store excess generation in the future. Therefore modeling of PHES systems should take into consideration wind power forecasts and thermal unit commitments as these interdependencies are fundamental to actual system operation.

A large body of work is available on the optimization of hydro-thermal systems [9]–[11] and stochastic optimization has been applied to hydro thermal problems in the past [12], [13]. The work presented in this paper is similar in concept however stochastic optimization is applied to the operation of a pumped storage plant in a power system with significant wind energy installation and as such historic wind power production profiles are used in place of historic hydro inflows to determine week-ahead targets for the storage.

Previous work in this area has generally focussed on the stochastic modeling and joint optimization of wind and hydro systems as in [14] and [15]. A number of papers have investigated the operation of PHES and large PHES in power systems with significant amounts of wind energy [16]–[20] where PHES is modeled deterministically. In operational mode PHES can either operate on a fixed cycle which is common [21] (i.e., refill the reservoir at the end of each day, week, etc.) or on a free cycle, similar to [20] where the value of storage is based on the marginal value of water held in the storage. These techniques may not be appropriate for larger PHES in systems with significant renewable generation for the following reasons: A fixed cycle takes no account of wind generation at the end of the cycle and,

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as shown later, may not exploit the storage to its full potential. Secondly in modeling a 'free cycle mode' a large storage will not attain its bounds within a short optimization period (for example 1–2 days) and the reservoir may empty within the first few simulation steps. Therefore for larger PHES systems the entire power system has to be optimized over longer periods such as a week, a month or a year. The question raised in this paper is how to deal with the inherent uncertainty in longer optimization periods in power systems with significant wind power production. In this paper therefore we are concerned with the derivation of robust end of day and end of week targets for the storage. PHES is free to 'react' to actual wind power production during the day, however it is constrained to finish each day at a predefined target as described in detail in Section III. It is the derivation of these daily and weekly targets that is the core goal of this paper.

III. Approach and Methodology

The methodology presented here specifically deals with closed-loop PHES (i.e., no hydro inflow) in power systems with high wind penetration. In this paper we built a model of a test power system and used it to solve medium and short-term PHES constraints in a two-step process to derive intertemporal targets for storage. Wind power production is the only stochastic variable. Although the model can also treat demand as a stochastic variable, demand is assumed here to be known with certainty as the focus is on the impacts of wind variability. This is a limitation of this analysis and should be borne in mind when considering results. Weekly end storage levels are derived using stochastic optimization and historic wind data. Daily end storage levels are derived using stochastic optimization and wind forecasts. Stochastic optimization is only applied to storage end levels. A detailed methodology is provided in Section III-C. Firstly the test system in presented in Section III-A and the modeling software is detailed in Section III-B.

A. Test System

The test system presented, while satisfying security of supply standards, is not intended to be an optimal portfolio, and is merely used as a test bench for modeling techniques. The test system modeled is that based on a potential portfolio for the All-Island system (AI) as in report [22]. Electricity demand for the island of Ireland in 2035 is assumed to be approximately 42 TWh with a peak of 7.4 GW. Generation units and unit capacity used are detailed in Table I. The system has one 500-MW PHES with 10 GWh of storage. One year of half hourly regional wind farm capacity factors (in time series format) from Ireland for the year 2008 were scaled up to match 2035 wind capacity values. This analysis does not consider detailed modeling of transmission issues, frequency and inertia issues of voltage stability. In this analysis however instantaneous wind penetration levels in the AI system are limited to 70% [23]. Pumped storage is modeled as 2 individual 250-MW generating and pumping units with a minimum stable generation level of 40 MW on each unit. The units are not allowed to pump and generate at the same time. Spinning reserve and replacement reserve are modeled. It is assumed that each PHES unit can contribute up to 100 MW to spinning reserve in generation mode and a full 250 MW in pumping mode. The demand for spinning reserve is set by the largest unit online with allowance for wind capacity as per [24].

TABLE I
Test Portfolio Characteristics

Type of Generation	Capacity AI (GW)	No of Units	Max Capacity of Units (MW)	Ramp rates MW/Min
Wind	8.4	-	-	-
CCGT	5.4	12	450	10
OCGT	2.0	20	100	6
Interconnector to Great Britain	2.0	4	500	-
Pumped Storage	0.5	2	250	20

B. PLEXOS Model

The software used throughout this work to solve the unit commitment and dispatch problem is PLEXOS [25], but the methodology can be implemented in any programmable platform. Full details of model equations are shown in the Appendix. PLEXOS is a power systems modeling tool used for electricity market modeling and planning worldwide [25]–[29] and recently in wind integration studies [30] and energy storage studies [31]. PLEXOS is a commercial modeling tool but is free for non-commercial research to academic institutions. This software was chosen because it is a flexible platform allowing user defined constraints. It is not a "black box" and the users can browse and verify the equations of the problem via a diagnostic tool. The PLEXOS modeling tool is used by the Commission for Energy Regulation (CER) in Ireland to validate Ireland's Single Electricity market and has a history of use in Ireland [32].

In this set-up PLEXOS co-optimizes hydro, thermal, renewable, and reserve classes and no heuristic or sequential approach is taken. Appendix A provides a description of the model equations but a short description is give here: modeling is carried out using mixed integer linear programming that aims to minimize an objective function subject to the expected cost of electricity dispatch and a number of constraints. The objective function of the model includes operational costs, consisting of fuel costs and carbon costs; a start up costs consisting of a fuel offtake at startup and a start cost. A penalty cost for unserved energy and a penalty cost for not meeting reserve requirements are also included in the objective function. Fuel consumption is calculated using a piecewise linear function as in [33]. System level constraints consist of an energy balance equation ensuring supply (net pumping demand) meets regional demand at each period. Water balance equations ensure water flow within the pumped storage units is conserved and tracked. Constraints on unit operation include minimum and maximum generation, maximum and minimum up and down time and ramp up and down rates. Constraints in relation to each unit's ability to provide reserve are also included. Ancillary services are optimized using techniques set out in [33]. In chronological

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TABLE II
OPTIMIZATION CONFIGURATION FOR EACH PHASE

Phase	Optimisation Interval	Optimisation Period	Optimisation Period 'Look ahead'	Optimisation Horizon
Step 1- Weekly targets: low resolution	6 hrs	1 year	0 days	1 year
Step 2 Daily targets: high resolution	2 hrs	1 day	6 days	1 year

mode PLEXOS solves for each period and maintains consistency across the full problem horizon. Temporal resolutions settings in relation to solving the problem are user defined and flexible. To avoid issues with intertemporal constraints at the simulation step boundaries a "*look ahead*" period is used. Look ahead means that the optimizer is given information about what happens ahead of the period of optimization and solves for this full period (i.e., simulation period + look ahead period) however only results for the simulation period are kept. In stochastic mode PLEXOS uses two-stage stochastic optimization. The classic two stage stochastic linear problem with recourse is described in [34].

In this analysis stochastic settings specifically force non-anticipativity on the PHES volume levels only to derive weekly and daily targets. The model is free to derive intra day levels. This means daily and weekly end volume levels for the PHES are identical for all scenarios of wind power production for these periods. Non anticipativity is not a hard constraint but there is a penalty for not meeting it in the objective function [(1) in the Appendix]. Softening this constraint allows the model to deviate from a unique volume in case of unserved energy. A description of the model formulation in linear programming format is provided in Appendix A. Note the same formulation is used in all steps of this process; time and optimization intervals change as per Table II described in the next section.

C. Proposed Methodology

A two step methodology or strategy for operation of the PHES is proposed which aims firstly to derive weekly and secondly daily end level storage targets for a large PHES from a system operator (as distinct from other market participants) point of view. PHES is free to operate in an optimal fashion within the day. The methodology also provides a technique for the 'bridging' of the daily and weekly targets.

The targets are derived from reservoir end volume levels (GWh) during model simulations of the complete power system. The decomposition of the targets from weekly to daily targets is based on wind power forecasts. The modeling process also considers forced outages, maintenance and ancillary services for all plant. Note non-anticipativity is forced on storage end volumes only and not on any other units. Step one of the process for the methodology uses low resolution optimization interval lengths over long optimization periods of time, while step two uses high resolution optimization interval lengths over shorter optimization periods of time. All model simulations are for a one year optimization horizon as detailed in Table II and

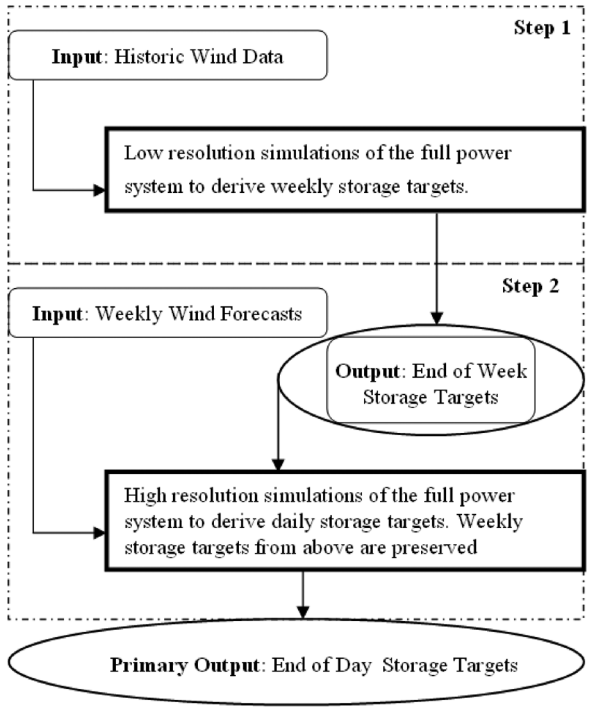


Fig. 1. Schematic of two step approach to development of targets for PHES in high wind systems.

Fig. 1. In Appendix A the model equations are shown. Step 1 of this methodology has 7 stochastic scenarios (S) of equal probability (k_s) simulated at an optimization interval (t) of 6 hours. Step 2 of this methodology has 3 stochastic scenarios (S) of equal probability (k_s) simulated at an optimization interval (t) of 2 hours. Note the model equations in the Appendix are the same for both step one and step two of this methodology.

1) Stochastic Method: Fig. 1 provides a schematic overview of the two step approach for the Stochastic Method.

Step one of the process aims to derive weekly targets (end of week) for the PHES storage. For large PHES storages, targets have to be derived in advance of knowing what the wind power production will be in the future. In reality, long range wind forecasts for periods up to a year are not available so in this analysis actual historic wind production data for the AI System is used to develop hourly annual profiles of wind power in time series format. 7 years of data are used from the year 2001 to 2007. The data was quality controlled and checked against local meteorological stations for consistency. It is assumed that each one of these 7 years of wind data represents a possible scenario of wind production for the target year. The scenarios are equally weighted. In this analysis discrete scenarios are used and it is assumed that the range of actual historic wind data presented captures and reflects the uncertainty inherent in annual wind production.

To derive the weekly targets one year of optimizations of the complete power system with optimization parameters set as per Table II (see Step 1) was undertaken across all historic years of wind data. These optimization settings were chosen so as to ensure the storage attained it bounds within the simulation horizon and also to make the program files manageable for the computer system. Without this annual look ahead, the system would

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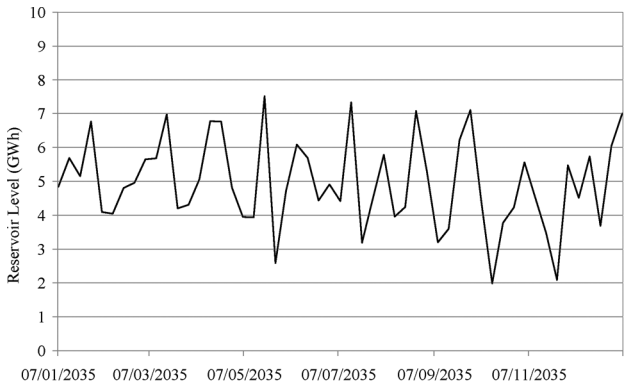


Fig. 2. Weekly target for PHES derived using stochastic method.

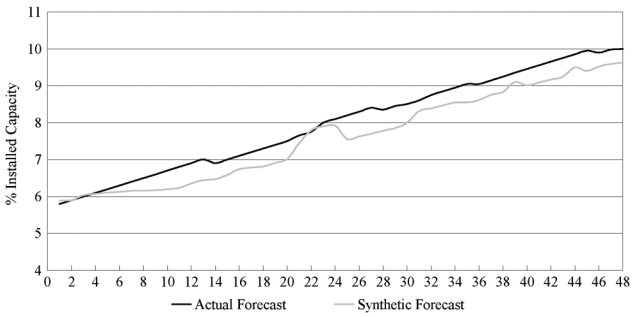


Fig. 3. Mean absolute error (%) of actual and synthetic forecast over a 48-hour period.

empty prematurely for large reservoirs. Within the simulation, stochastic settings for storage force trajectory non-anticipativity as per (13) in Appendix A. This provides one unique weekly storage trajectory that is optimal considering all seven years of wind data. From the solution database the end of week storage levels were extracted and retained for step 2 of the process. Fig. 2 shows the weekly targets derived using this method.

Step two of the methodology aims to derive daily targets for PHES while preserving the weekly targets determined in step one. Daily storage targets should be set so as to give PHES adequate flexibility to deal with daily variability in the power system but also ensure that PHES finishes each day in a position to manage the next day's and number of day's variability and unpredictability of wind.

Daily storage targets for each day of the week are developed in the context of weekly wind power forecasts from the year 2008. In this analysis 3 individual 7 day wind energy forecasts are used. The forecast error is simulated by Auto Regressive Moving Average (ARMA) series similar to [35] that are established by tuning the statistical characteristics to those of actual 5 day forecasts supplied by Eirgrid. This method of wind forecast simulation may overestimate the variability of wind profiles due to the lack of smoothing that occurs when wind capacity is increased and is a limiting factor in the study. The development of the forecast error over a 48-hour period is shown in Fig. 3. The graph shows the mean absolute error (as % of installed wind capacity) for a generated forecast and an actual forecast. After the 48-hour period the error is assumed to be constant.

In Step 2 the full power system was again optimized forcing the end volume levels for the pumped storage reservoir to be

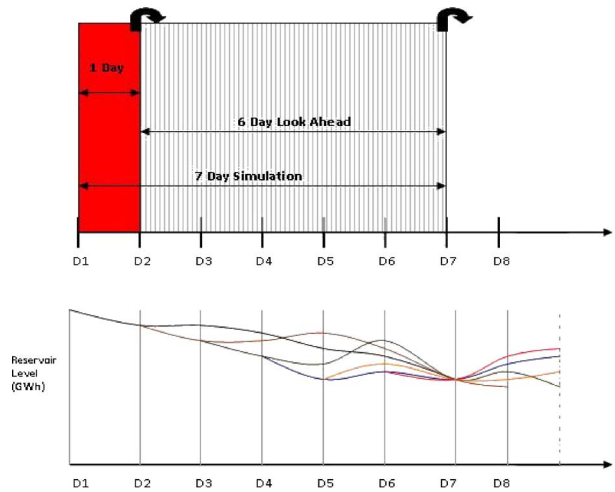


Fig. 4. Schematic of "look ahead" period on top with rolling reservoir levels on bottom. X-axis is days of simulation.

the same across the full range of wind forecast scenarios for the end of each day while also "hitting" the discrete end of week targets as developed in step 1. As per Table II a year of daily optimizations with a 6 day look ahead is used. This ensures the optimization can "see" the end of week storage target and rolls forward towards it. End volume levels are calculated for 7 days (1 optimization day + 6 days of look ahead) at a time but only the first day of the simulation is kept and targets are recalculated again for each day as the simulation "rolls" forward while end of week targets developed from Step 1 of this methodology are enforced. This is shown in schematic form in Fig. 4.

End of day targets were extracted and written to a text file. These targets were tested in a final deterministic simulation and compared to a conventional *Refill Method* as described next.

All simulation were executed on a workstation with an Intel Xeon 3.0-GHz processors with 12 GB of RAM running Microsoft Windows 7 with 64 bit and the Xpress MP MIP [36] solver with a relative optimality criterion of 0.2%. The longest stochastic simulation was 27 hours.

2) Weekly Refill Method: A simple weekly "*Refill*" method is also put forward as a means for comparison. This method is put forward as many storage operators balance large reservoirs on a weekly basis [21]. In this method the reservoir is simply returned to its full level at the end of each week. End of day storage targets are set as the simple average of individual deterministic simulations of each day ahead wind forecasts.

IV. Results

To test and compare the methods on economic terms, the power system was simulated deterministically for one year (365 optimization periods) at hourly intervals. Daily targets as derived above for the 2 methods were enforced in all simulations. This was done to replicate a "real" operational run. Instead of wind forecasts, the actual realized wind was used from the year 2008. All model simulations have identical parameters (except end of day storage targets) including the times and duration of maintenance and forced outages so it is assumed that the cause of any difference in economic output is attributed to storage levels and the way the storage levels are derived.

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TABLE III
RELATIVE INCREASE IN ANNUAL TOTAL SYSTEM COSTS RELATIVE TO PERFECT FORESIGHT CASE (%)

Stochastic Method	Refill Method
+0.12%	+0.28%

TABLE IV
ANNUAL GENERATION (GWH) FOR AI UNITS FOR FLEXIBLE INTERCONNECTOR FLOW SCENARIO

	Refill Method	Stochastic Method	Perfect Foresight
OCGT Plant	191	181	170
Pumped Storage	435	452	408
CCGT Plant	20138	19967	19919
Renewables	23042	23214	23331
Pump Load	580	602	544
Import to AI	2882	3387	2824
Export from AI	3824	4113	3623
Demand	42285	42486	42485
Wind Curtailment (%)	2.04%	1.31%	0.82%
Total Annual Generation Costs (M €)	€2,104	€2,101	€2,098

Table III compares the total annual generation cost for a "real" operational model simulation for both methods compared to a perfect foresight model simulation where wind production is known exactly. The flow across the interconnector is assumed to be flexible in this simulation. Total generation cost includes fuel cost, variable operation and maintenance costs, start and shutdown costs and emissions costs.

While the difference in % terms is small, the total annual system operation cost for the simulation is high, (approximately 2 billion Euro for this system), thus the annual savings that the Stochastic Method would provide over the Refill Method are approximately 3.3 million Euro. The Stochastic Method is a more sophisticated strategy of operation than the conventional strategy; however its implementation cost is not excessive other than the cost of including stochastic programming. All other resources, such as historic wind profiles and wind forecast should be available to a system operator.

Table IV compares the total annual generation for each generation category for a scenario where flow on the interconnector to GB is flexible. Table V compares the annual generation for each generation category in the AI system for a scenario where the flow on the interconnector is fixed day ahead based on a wind forecast.

Looking firstly at the flexible IC flow scenario (Table IV), it can be seen that when uncertainty is accounted for in wind power forecasts the generation level for thermal generation increases relative to a perfect foresight case. Total annual system costs is lowest (as expected) for the perfect foresight however

TABLE V
ANNUAL GENERATION (GWH) FOR AI UNITS FOR FIXED INTERCONNECTOR FLOW SCENARIO

Category	Refill Method	Stochastic Method	Perfect Foresight
OCGT Plant	527	365	286
Pumped Storage	472	662	584
CCGT Plant	22677	22891	22922
Renewables	21229	21241	21263
Pump Load	630	883	779
Import to AI	2327	2327	2327
Export from AI	4118	4118	4118
Customer Load	42484	42485	42485
Wind Curtailment (%)	9.75%	9.70%	9.61%
Total Annual Generation Costs (M €)	€2,148	€2,146	€2,140

the Stochastic Method for deriving PHES targets provides lower costs that the Refill Method. One of the reasons for this is that the Stochastic Method better exploits interconnector import capacity (imports into the AI system are higher for the stochastic optimization method) as the storage end level is co-optimized in association with interconnection across all scenarios of wind forecasts leading to better coordination of interconnection and storage strategies. This is not surprising as pumped storage and interconnection have similar flexible characteristics within the power system and should "compete" closely to provide flexibility. The operation of pumped storage under uncertainty also leads to a higher capacity factor compared to a perfect foresight operation and also reduced wind curtailment when compared to the Refill Method.

Looking at the scenario when interconnector flow is fixed and is scheduled day ahead (Table V), it is shown again stochastic optimization of PHES targets provides the lowest increase in costs relative to a perfect foresight determination of PHES targets. When interconnector flows are fixed the magnitude of the savings are greater. This is because the interconnector is no longer allowed to provide the same level of flexibility. In the Stochastic Method, the capacity factor of the PHES is increased and the model exploits CCGT units more often whereas the Refill Method employs less CCGT production and calls on OCGT units more often thus leading to higher total system costs.

A weekly dispatch of generation for the AI system under a fixed IC flow scenario for both the Stochastic Method and Refill Method is shown in Figs. 5 and 6. Imports and exports are the same for both scenarios and are not shown. It can be seen that the daily and weekly targets derived by the Stochastic Method allow greater flexibility in PHES operation allowing it to pump in the early week when wind generation is high, thus reducing wind curtailment and system costs and also enabling it to generate in the latter part of the week.

Fig. 6 shows the same week and IC scenario for the Refill Method. When the storage reservoir is recycled each week it limits the flexibility of storage to pump in times of high wind

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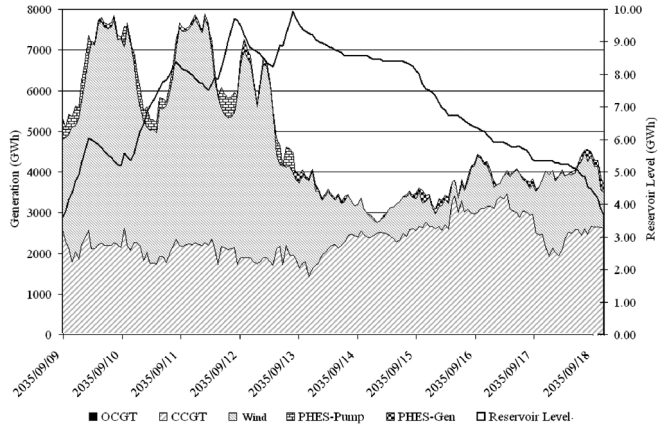


Fig. 5. Generation stack for a week for fixed IC scenario and Stochastic Method for deriving PHES targets. Generation levels are shown on left y-axis. Reservoir levels are shown on right y-axis.

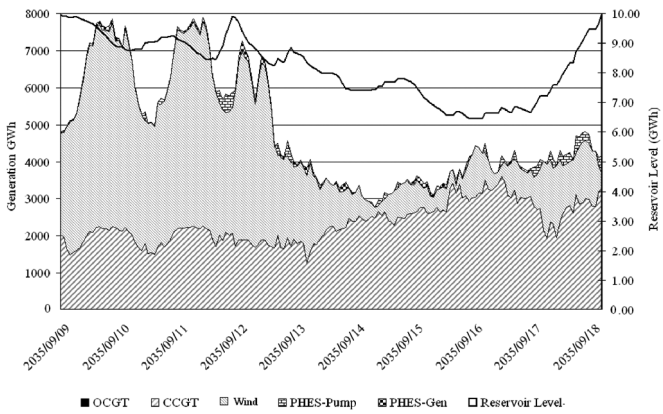


Fig. 6. Generation stack for a week for fixed IC scenario and Refill Method for deriving PHES targets. Generation levels are shown on left y-axis. Reservoir levels are shown on right y-axis.

production. The storage is also constrained to recycle back to its full level at the end of the week when wind generation is lower. This method would be suitable for thermal generation dominated systems but in this case, where there is significant wind generation it can be seen to be a poor operational strategy.

1) Sensitivity Analysis: To determine the sensitivity of total system costs to a variation in wind power profiles a number of extra simulations were undertaken. The simulations followed the same methodology as describe above, however each of the individual historic 7 years of wind data was used to develop the wind forecasts thus testing the sensitivity of results to each wind profile. Table VI presents results of this analysis.

Results of the sensitivity analysis show that the Stochastic Method provides lower total system costs and on average produce costs that are 0.19% higher than a perfect forecast determination of targets compared to a conventional Refill Method which are on average 0.38% higher for a flexible IC scenario.

V. Conclusion

This paper has described and presented a methodology for deriving intertemporal targets for pumped hydro energy storage units that need to be optimized over long periods of time due to their large energy capacity. The methodology allows for the fact that these decisions have to be made in the context of uncertainty and stochastic optimization is presented as a means of deriving end targets for storage in the context of wind forecast uncertainty. The paper shows that daily and weekly end volumes for PHES derived using stochastic optimization provide lower total system generation cost than a conventional Refill Method. This is primarily because (in cases where interconnection scheduling is flexible) the stochastic optimization method is better able to coordinate the efficient operation of the PHES units and interconnector to provide lowest operational cost. In cases where interconnector flow are scheduled day ahead the Stochastic Method better exploits cheaper generation such as CCGT units to provide corrective adjustment to deviations from the expected wind forecast.

TABLE VI
Sensitivity Analysis Showing Relative Increase (%) in Annual Total System Costs Relative to Perfect Foresight Case for Fixed and Flexible Interconnector Flows

Flexible IC	Stochastic Method	Refill Method
Min	0.03%	0.07%
Max	0.48%	0.81%
Average	0.19%	0.38%
Fixed IC Flow		
Min	0.34%	0.96%
Max	0.97%	1.69%
Average	0.58%	1.15%

Appendix A
PLEXOS Equations

Indices

j	Generation unit.
J	Total number of generators.
t	Optimization interval.
T	Optimization horizon.
stor	Index related specifically to pumped storage unit.
s	Stochastic scenario.
S	Total number of scenarios.
$\mathrm{RES}^{\mathrm{up}}$	Upper storage reservoir.
$\mathrm{RES}_{\mathrm{low}}$	Lower storage reservoir.

Variables

V_{jt}	Integer on/off decision variable for unit j at period t.
X_{jt}	Integer on/off decision variable for pumped storage pumping unit j at period t.
U_{jt}	Variable that $=1$ at period t if unit j has started in previous period else 0.

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P_{jt}	Power output of unit j (MW).
H_{jt}	Pump load for unit j period t (MW).
W_{int}	Flow into reservoir at time t (MWh).
W_{outt}	Flow out of reservoir at time t (MWh).
W_t	Volume of storage at a time t (MWh).

Parameters

vl	Penalty for loss of load (€/MWh).
use	Unserved energy (MWh).
vs	Penalty for reserve not met.
usr	Reserve not met (MWh).
D	Demand (MW).
obj	Objective function.
k_s	Probability of scenario s.
n_{jt}	No load cost unit j in period t (€).
c_{jt}	Start cost unit j in period t (€).
m_{jt}	Production cost unit j in period t (€).
e_{stor}	Efficiency of pumping unit (%).
$pmax_j$	Max power output of a unit j (MW).
$pmin_j$	Mini stable generation of unit j (MW).
$pmpmax_{stor}$	Max pumping capacity of pumping unit.
J_j	Available units in each generator.
J_{stor}	Number of pumping units.
MRU_j	Maximum ramp up rate (MW/min).
MRD_j	Maximum ramp down rate (MW/min).
MUT_j	Minimum up time (hours).
W_{INT}	Initial volume of reservoir (GWh).
W	Maximum volume of storage (GWH).
δ	Penalty price if storage end volume for upper reservoir deviates from unique solution-penalty for non-respecting non-anticipativity. (€).
θ_t	Deviation of storage level from a unique solution at period t. Measure of non-compliance with non-anticipativity. (MWh).

Objective Function:

$$OBJ = Min \sum_{\substack{\forall s\in S\\ \forall t\in T\\ \forall j\in J}} k_s(c_{jt}.U_{jt} + n_{jt}V_{jt} + m_{jt}.P_{jt} + vl.use_t + vs.usr_t + \delta\theta_t). \quad (1)$$

The objective function in PLEXOS is to minimize the start up cost of each unit [start cost (€)* number of starts of a unit] + the no load cost of each online unit + production costs of each online unit + the penalty for unserved load + the penalty of unserved reserve. The last terms in the objective function deals with the cost of non compliance for non-anticipativity for storage levels as set out in (13). The penalty is set to a high value so in general the model will meet the non-anticipativity unless there is risk of very high prices or unserved energy. The objective function is minimized within each simulation period. The simulation solution must also satisfy the constraints below:

Energy Balance Equation:

$$\sum_{s\in S}\sum_{t\in T} P_{jst} - H_{jst} + use_t = D_{ts}. \quad (2)$$

Energy balance equation states that the power output from each unit at each interval minus the pump load from pumped storage units for each interval + unserved energy must equal the demand for power at each interval. As the penalty for unserved energy is high and part of the objective function, the model will generally try to meet demand.

Operation Constraints on Units: Basic operational constraints that limit the operation and flexibility of units such as maximum generation, minimum stable generation, minimum up/down times and ramp rates:

$$-V_{jst} + U_{jst} \geq -1 \quad \forall t = 1 \quad (3)$$

$$V_{jst} - V_{jst+1} + U_{jst+1} \geq 0. \quad (4)$$

These two equations define the start definition of each unit and are used to track the on/off status of units:

$$P_{jst} - P\max_j .V_{jst} \leq 0. \quad (5)$$

A generation units power output cannot be greater than its maximum capacity:

$$P_{jst} - P\min_j .V_{jst} \geq 0. \quad (6)$$

A generation units output must be greater than its minimum stable generation when the unit is online:

$$H_{jst} - Pmp\max_{Stor} .X_{jst} \leq 0. \quad (7)$$

Pumping load must less than maximum pumping capacity for each pumping unit:

$$V_{jst} + X_{jst} \leq 1 \text{ where } j \in stor. \quad (8)$$

This constraint limits a pumped storage unit from pumping and generating at same time:

$$A_{pjst} \geq V_{jst} - V_{jst-1} \; \forall t..t - MUT_{js} - 1$$
$$V_{jst} \geq A_{pjst} - \sum_{t-MUT_{js}+1}^{t} V_{jst}/MUT_{js} \; \forall t. \quad (9)$$

Minimum Up Times:[1] (Note the following text is directly from the PLEXOS Help files). The variable A_p tracks if any starts have occurred on the unit inside the periods preceding p with a window equal to MUT. In other words, if no starts happen in the last MUT periods, then A_p will be zero, but if one (or more) starts have occurred, then A_p will equal unity. The MUT constraints then sets a lower bound on the unit commitment that

[1]PLEXOS Help Files

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is normally below zero, but when a unit is started, the bound rises above zero until the minimum up time has expired. This fractional lower bound when considered in an integer program forces the unit to stay on for its minimum up time:

$$A_{pjst} \geq V_{jst-1} - V_{jst} \; \forall t..t - MDT_{js} + 1$$
$$V_{jst} \leq 1 + \sum_{t}^{t-MDT_{js}+1} V_{jst}/MDT_j - A_{pjst} \; \forall t.$$

Minimum Down Times: The variable A_p tracks if any units have been shut down inside the periods preceding p with a window equal to MDT. In other words, if no units are shutdown in the last MDT periods, then A_p will be zero, but if one (or more) shutdown, then A_p will equal unity. The MDT constraints then set an upper bound on the unit commitment that is normally above unity, but when a unit is stopped, the bound falls below unity until the minimum down time has expired:

$$P_{jst} - P_{js.t-1} - MRU_j.V_{jst} - p_{\min_j}.U_j \leq 0 \quad (10)$$
$$p_{\min j}.P_{jst} + P_{jst} - P_{js.t-1} - P_{jst}.(MRD_j - p_{\min_j}) \leq 0. \quad (11)$$

Maximum Ramp up and down constraints: These constraints limit the change in power output from one time period to another.

Water Balance Equations: These equations track the passage of water from the lower reservoir to the upper reservoir. In this set-up there is no inflow and water volume is conserved:

$$W_{stR} + W_{out.stR} - W_{in.stR} = W_{INT,R} \forall t = 1, R \in RES_{Up}, RES_{low}$$
$$W_{st.RES^{up}} + W_{out,s,RES^{up}} - W_{in,s,RES^{up}} = 0$$
$$e_{stor}.H_{jst.RES^{up}} - W_{in.stRES^{up}} = 0$$
$$P_{stor.st} - W_{out.st.RES^{up}} = 0 \quad (12)$$
$$- W_{s_n tRES^{up}} + W_{s_n t+1 RES^{up}} + \theta_{st} = 0 \quad \forall n \in S. \quad (13)$$

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